THE ESSENTIAL REVIEW

Grade 10 Advanced

مراجعة مركّزة من الكتاب

Algebra. Similarity. Trigonometry.

25 learning outcomes

100 textbook questions

60 minute online mock

Name / الاسم _____ Class / الصف _____

A short route to confident practice

Read the rule → solve without looking at the key → check your steps → repeat the topic online.

اقرأ القاعدة، حلّ بنفسك، ثم راجع خطواتك. لكل مخرج 4 أسئلة مختارة، الرسومات من الكتاب الأصلي.

Your topic map

Q1	Polynomial operations	2
Q2	FOIL and binomial products	2
Q3	Factoring by GCF and grouping	3
Q4	Factoring monic trinomials	3
Q5	Interior angles of polygons	4
Q6	Properties of parallelograms	5
Q7	Tests for parallelograms	6
Q8	Kites	7
Q9	Dilation coordinates	8
Q10	Similar polygons	9
Q11	Dilation scale factors	10
Q12	AA triangle similarity	11
Q13	Proportional parts in triangles	12
Q14	Parts of similar triangles	13
Q15	Geometric mean	14
Q16	Pythagorean theorem and converse	15
Q17	Special right triangles	16
Q18	Right-triangle trigonometry	17
Q19	Angles of elevation and depression	18
Q20	Area with two sides and an angle	19
Q21	Special-product factorization	20
Q22	Interior and exterior angle reasoning	21
Q23	Solving with similar triangles	22
Q24	Distance and midpoint in space	24
Q25	Law of Sines and the ambiguous case	26
KEY	MCQ answers	28
KEY	Worked written answers	29

Selected from the uploaded Reveal textbook, printed pp. 57–220 and 627–686, aligned to the 25 established learning outcomes. Numbers and diagrams come from the book; MCQ choices and worked solutions are added. This is a study selection, not an official exam or a prediction. Use the rounding instructions stated in each original book question. The website mock is a separate 60-minute practice format.

الأسئلة من الكتاب فقط، مع توثيق الصفحة والتمرين. لا يمكن ضمان ورود أسئلة بعينها في الاختبار النهائي.

01 / CHOOSE & CHECK

Q1 — Q2

Q1 Polynomial operations
عمليات كثيرات الحدود

Remember Order terms by decreasing powers. Combine like terms. When subtracting a polynomial, change the sign of every term in the second polynomial.

1.1 MCQ

Write in standard form and identify the leading coefficient.

$$-y^3+3y-3y^2+2$$

Reveal · p. 635 · 10 · MCQ adaptation

A $-y^3-3y^2+3y+2,\quad 1$
B $-y^3-3y^2+3y+2,\quad 2$
C $-y^3-3y^2+3y+2,\quad -1$
D $y^3+3y^2-3y-2,\quad 1$

1.2 MCQ

Write in standard form and identify the leading coefficient.

$$\frac{1}{2}x-3x^4+7$$

Reveal · p. 635 · 13 · MCQ adaptation

A $-3x^4+\frac{x}{2}+7,\quad 2$
B $-3x^4+\frac{x}{2}+7,\quad -3$
C $3x^4-\frac{x}{2}-7,\quad 3$
D $-3x^4+\frac{x}{2}+7,\quad 3$

1.3 MCQ

Find the sum or difference.

$$(2h^2-5h)+(7h-3h^2)$$

Reveal · p. 635 · 22 · MCQ adaptation

A $-h^2+2h$
B $-h^2+2h+1$
C h^2-2h
D $-h^2+2h-x$

1.4 MCQ

Find the sum or difference.

$$(2x+3x^2)-(7-8x^2)$$

Reveal · p. 635 · 26 · MCQ adaptation

A $11x^2+2x-6$
B $-11x^2-2x+7$
C $11x^2+x-7$
D $11x^2+2x-7$

Q2 FOIL and binomial products
ضرب ثنائيي الحد بطريقة FOIL

Remember (a + b)(c + d) = ac + ad + bc + bd. Combine the two middle products when they are like terms.

2.1 MCQ

Find the product.

$$(c+3)(3c-5)$$

Reveal · p. 649 · 1 · MCQ adaptation

A $3c^2+4c-14$
B $3c^2+4c-15$
C $3c^2+5c-15$
D $3c^2+3c-15$

2.2 MCQ

Find the product.

$$(g+10)(2g-5)$$

Reveal · p. 649 · 2 · MCQ adaptation

A $2g^2+15g-50$
B $2g^2+16g-50$
C $2g^2+14g-50$
D $2g^2+15g-49$

2.3 MCQ

Find the product.

$$(5a+3)(6a+5)$$

Reveal · p. 649 · 3 · MCQ adaptation

A $30a^2+44a+15$
B $30a^2+42a+15$
C $30a^2+43a+16$
D $30a^2+43a+15$

2.4 MCQ

Find the product.

$$(4x+1)(6x+3)$$

Reveal · p. 649 · 4 · MCQ adaptation

A $24x^2+17x+3$
B $24x^2+18x+4$
C $24x^2+18x+3$
D $24x^2+19x+3$

01 / CHOOSE & CHECK

Q3 — Q4

Q3 Factoring by GCF and grouping

التحليل بالعامل المشترك والتجميع

Remember Factor out the greatest common factor first. For grouping, arrange terms so both groups contain the same binomial factor.

3.1 MCQ

Factor completely using the Distributive Property.

$$fg + 4f - 5g - 20$$

Reveal · p. 665 · 11 · MCQ adaptation

A $(f-5)(g+4)$
B $-(f-5)(g+4)$
C $2(f-5)(g+4)$
D $(f-4)(g+4)$

3.2 MCQ

Factor completely using the Distributive Property.

$$hj - 2h + 5j - 10$$

Reveal · p. 665 · 13 · MCQ adaptation

A $-(h+5)(j-2)$
B $2(h+5)(j-2)$
C $(h+6)(j-2)$
D $(h+5)(j-2)$

3.3 MCQ

Factor completely using the Distributive Property.

$$xy - 2x + y - 2$$

Reveal · p. 665 · 14 · MCQ adaptation

A $2(x+1)(y-2)$
B $(x+2)(y-2)$
C $(x+1)(y-2)$
D $-(x+1)(y-2)$

3.4 MCQ

Factor completely using the Distributive Property.

$$pv + 8p + 12v + 96$$

Reveal · p. 665 · 20 · MCQ adaptation

A $(p+13)(v+8)$
B $(p+12)(v+8)$
C $-(p+12)(v+8)$
D $2(p+12)(v+8)$

Q4 Factoring monic trinomials

تحليل ثلاثيات الحدود ذات المعامل الرئيسي 1

Remember Find integers m and n with m + n = b and mn = c. Then $x^2 + bx + c = (x + m)(x + n)$.

4.1 MCQ

Factor using integers.

$$x^2 + 17x + 42$$

Reveal · p. 671 · 1 · MCQ adaptation

A $-(x+3)(x+14)$
B $2(x+3)(x+14)$
C $(x+4)(x+15)$
D $(x+3)(x+14)$

4.2 MCQ

Factor using integers.

$$y^2 - 17y + 72$$

Reveal · p. 671 · 2 · MCQ adaptation

A $2(y-9)(y-8)$
B $(y-8)(y-7)$
C $(y-9)(y-8)$
D $-(y-9)(y-8)$

4.3 MCQ

Factor using integers.

$$a^2 + 8a - 48$$

Reveal · p. 671 · 3 · MCQ adaptation

A $(a-3)(a+13)$
B $(a-4)(a+12)$
C $-(a-4)(a+12)$
D $2(a-4)(a+12)$

4.4 MCQ

Factor using integers.

$$n^2 - 2n - 35$$

Reveal · p. 671 · 4 · MCQ adaptation

A $(n-7)(n+5)$
B $-(n-7)(n+5)$
C $2(n-7)(n+5)$
D $(n-6)(n+6)$

01 / CHOOSE & CHECK

Q5

Q5 Interior angles of polygons
الزوايا الداخلية للمضلعات

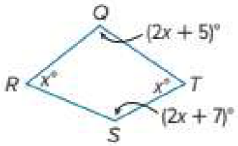
Remember For a simple n-sided polygon, the interior angle sum is (n − 2)180°. For a regular polygon, each interior angle is (n − 2)180°/n.

5.1 MCQ

Find the measure of each interior angle. Options list the vertices in the stated order.

Reveal · p. 63 · 1 · MCQ adaptation

1.



A $(Q, R, S, T) = (117, 56, 119, 56)^\circ$

B $(Q, R, S, T) = (125, 60, 127, 60)^\circ$

C $(Q, R, S, T) = (121, 58, 123, 58)^\circ$

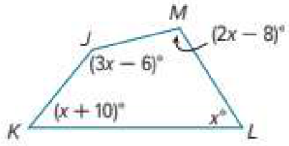
D $(Q, R, S, T) = (120, 60, 120, 60)^\circ$

5.2 MCQ

Find the measure of each interior angle. Options list the vertices in the stated order.

Reveal · p. 63 · 2 · MCQ adaptation

2.



A $(J, K, L, M) = (150, 52, 62, 96)^\circ$

B $(J, K, L, M) = (150, 62, 52, 96)^\circ$

C $(J, K, L, M) = (144, 60, 50, 92)^\circ$

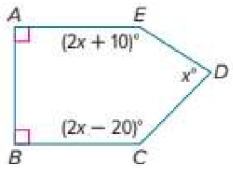
D $(J, K, L, M) = (156, 64, 54, 100)^\circ$

5.3 MCQ

Find the measure of each interior angle. Options list the vertices in the stated order.

Reveal · p. 63 · 3 · MCQ adaptation

3.



A $(A, B, C, D, E) = (90, 90, 128, 74, 158)^\circ$

B $(A, B, C, D, E) = (90, 90, 140, 80, 170)^\circ$

C $(A, B, C, D, E) = (90, 90, 124, 72, 154)^\circ$

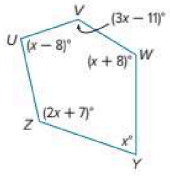
D $(A, B, C, D, E) = (90, 90, 158, 74, 128)^\circ$

5.4 MCQ

Find the measure of each interior angle. Options list the vertices in the stated order.

Reveal · p. 63 · 4 · MCQ adaptation

4.



A $(U, V, W, Y, Z) = (62, 199, 78, 70, 147)^\circ$

B $(U, V, W, Y, Z) = (68, 217, 84, 76, 159)^\circ$

C $(U, V, W, Y, Z) = (60, 143, 76, 68, 193)^\circ$

D $(U, V, W, Y, Z) = (60, 193, 76, 68, 143)^\circ$

01 / CHOOSE & CHECK

Q6

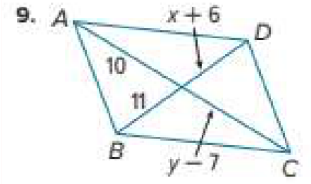
Q6 Properties of parallelograms
خصائص متوازيات الأضلاع

Remember Opposite sides and opposite angles are congruent. Consecutive angles are supplementary. A diagonal divides a parallelogram into two congruent triangles.

6.1 MCQ

Find each variable in the parallelogram.

Reveal · p. 72 · 9 · MCQ adaptation

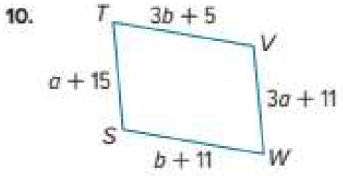


A $(x, y) = (16, 3)$
B $(x, y) = (5, 17)$
C $(x, y) = (11, 10)$
D $(x, y) = (17, 5)$

6.2 MCQ

Find each variable in the parallelogram.

Reveal · p. 72 · 10 · MCQ adaptation

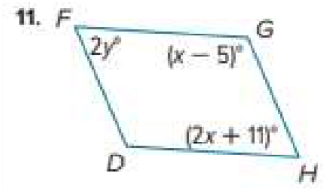


A $(a, b) = (2, 3)$
B $(a, b) = (3, 2)$
C $(a, b) = (13, 6)$
D $(a, b) = (4, 3)$

6.3 MCQ

Find each variable in the parallelogram.

Reveal · p. 72 · 11 · MCQ adaptation

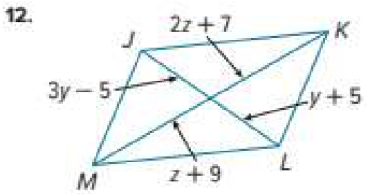


A $(x, y) = (58, 29)$
B $(x, y) = (60, 60)$
C $(x, y) = (53, 58)$
D $(x, y) = (58, \frac{127}{2})$

6.4 MCQ

Find each variable in the parallelogram.

Reveal · p. 72 · 12 · MCQ adaptation



A $(y, z) = (3, 2)$
B $(y, z) = (5, 9)$
C $(y, z) = (5, 2)$
D $(y, z) = (2, 5)$

01 / CHOOSE & CHECK

Q7

Q7 Tests for parallelograms
اختبارات متوازي الأضلاع

Remember Sufficient tests include both pairs of opposite sides congruent, both pairs of opposite angles congruent, diagonals bisecting each other, or one pair of opposite sides both parallel and congruent.

7.1 MCQ

Determine whether the markings guarantee a parallelogram. Choose the correct justification.

Reveal · p. 79 · 1 · MCQ adaptation

1.



A Yes: one pair of opposite sides is both parallel and congruent.

B Yes: the diagonals are marked perpendicular.

C No: a parallelogram cannot have these markings.

D Yes: any one pair of congruent sides is sufficient.

7.2 MCQ

Determine whether the markings guarantee a parallelogram. Choose the correct justification.

Reveal · p. 79 · 2 · MCQ adaptation

2.



A Yes: the diagonals are marked perpendicular.

B No: a parallelogram cannot have these markings.

C Yes: any one pair of congruent sides is sufficient.

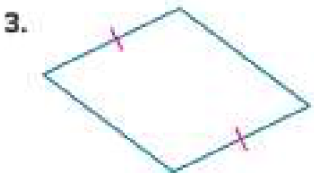
D Yes: both pairs of opposite angles are congruent.

7.3 MCQ

Determine whether the markings guarantee a parallelogram. Choose the correct justification.

Reveal · p. 79 · 3 · MCQ adaptation

3.



A No: a parallelogram cannot have these markings.

B Yes: any one pair of congruent sides is sufficient.

C Not enough information: only one pair of opposite sides is congruent.

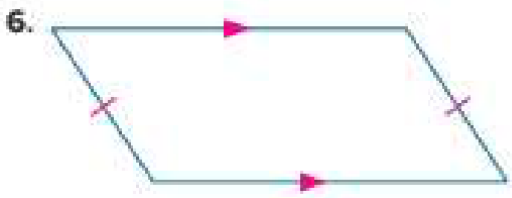
D Yes: the diagonals are marked perpendicular.

7.4 MCQ

Determine whether the markings guarantee a parallelogram. Choose the correct justification.

Reveal · p. 79 · 6 · MCQ adaptation

6.



A Yes: any one pair of congruent sides is sufficient.

B Not enough information: one pair is parallel and the other pair is congruent.

C Yes: the diagonals are marked perpendicular.

D No: a parallelogram cannot have these markings.

01 / CHOOSE & CHECK

Q8

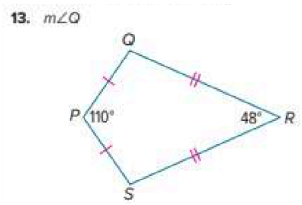
Q8 Kites
الطائرات الورقية

Remember In a kite with AB = AD and CB = CD, AC is the symmetry diagonal. AC is perpendicular to BD and bisects BD. The opposite angles at B and D are congruent.

8.1 MCQ

Find the requested angle measure in the kite.

Reveal · p. 106 · 13 · MCQ adaptation

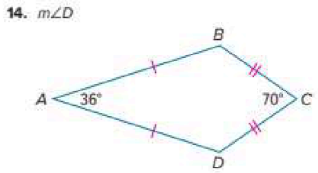


A 103
B 79
C 99
D 101

8.2 MCQ

Find the requested angle measure in the kite.

Reveal · p. 106 · 14 · MCQ adaptation

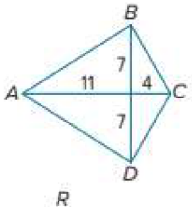


A 53
B 125
C 127
D 129

8.3 MCQ

Find the requested side in simplest radical form, then the perimeter to the nearest tenth.

Reveal · p. 106 · 15 · MCQ adaptation

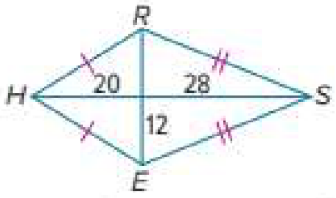


A $BC = \sqrt{65}, \quad P \approx 58.5$
B $BC = \sqrt{65}, \quad P \approx 42.2$
C $BC = \sqrt{170}, \quad P \approx 42.2$
D $BC = \sqrt{65}, \quad P \approx 21.1$

8.4 MCQ

Find the requested side in simplest radical form, then the perimeter to the nearest tenth.

Reveal · p. 106 · 16 · MCQ adaptation



A $RH = 4\sqrt{34}, \quad P \approx 107.6$
B $RH = 4\sqrt{58}, \quad P \approx 107.6$
C $RH = 4\sqrt{34}, \quad P \approx 53.8$
D $RH = 4\sqrt{34}, \quad P \approx 96.0$

01 / CHOOSE & CHECK

Q9

Q9

Dilation coordinates

إحداثيات التمدد

Remember For a dilation centered at the origin, (x, y) maps to (kx, ky). Use image/original to find k, retaining the signs of coordinates.

9.1 MCQ

Graph the image after a dilation centred at the origin. Choose the complete set of image vertices.

$$F(-10,4) \quad G(-4,4) \quad H(-4,-8)$$

$$k = \frac{1}{4}$$

Reveal · p. 120 · 13 · MCQ adaptation

A
$F' = (\frac{5}{2}, -1)$
$G' = (1, -1)$
$H' = (1, 2)$

B
$F' = (-\frac{25}{2}, 5)$
$G' = (-5, 5)$
$H' = (-5, -10)$

C
$F' = (-\frac{5}{2}, 1)$
$G' = (-1, 1)$
$H' = (-1, -2)$

D
$F' = (-40, 16)$
$G' = (-16, 16)$
$H' = (-16, -32)$

9.2 MCQ

Graph the image after a dilation centred at the origin. Choose the complete set of image vertices.

$$X(2,-1) \quad Y(-6,4) \quad Z(-2,-5)$$

$$k = \frac{5}{4}$$

Reveal · p. 120 · 14 · MCQ adaptation

A
$X' = (\frac{9}{2}, -\frac{9}{4})$
$Y' = (-\frac{27}{2}, 9)$
$Z' = (-\frac{9}{2}, -\frac{45}{4})$

B
$X' = (\frac{5}{2}, -\frac{5}{4})$
$Y' = (-\frac{15}{2}, 5)$
$Z' = (-\frac{5}{2}, -\frac{25}{4})$

C
$X' = (\frac{8}{5}, -\frac{4}{5})$
$Y' = (-\frac{24}{5}, \frac{16}{5})$
$Z' = (-\frac{8}{5}, -4)$

D
$X' = (-\frac{5}{2}, \frac{5}{4})$
$Y' = (\frac{15}{2}, -5)$
$Z' = (\frac{5}{2}, \frac{25}{4})$

9.3 MCQ

Graph the image after a dilation centred at the origin. Choose the complete set of image vertices.

$$M(4,6) \quad N(-6,2) \quad P(0,-8)$$

$$k = \frac{3}{4}$$

Reveal · p. 120 · 15 · MCQ adaptation

A
$M' = (3, \frac{9}{2})$
$N' = (-\frac{9}{2}, \frac{3}{2})$
$P' = (0, -6)$

B
$M' = (\frac{16}{3}, 8)$
$N' = (-8, \frac{8}{3})$
$P' = (0, -\frac{32}{3})$

C
$M' = (-3, -\frac{9}{2})$
$N' = (\frac{9}{2}, -\frac{3}{2})$
$P' = (0, 6)$

D
$M' = (7, \frac{21}{2})$
$N' = (-\frac{21}{2}, \frac{7}{2})$
$P' = (0, -14)$

9.4 MCQ

Graph the image after a dilation centred at the origin. Choose the complete set of image vertices.

$$R(-2,6) \quad S(0,-1) \quad T(-5,3)$$

$$k = \frac{3}{2}$$

Reveal · p. 120 · 16 · MCQ adaptation

A
$R' = (-\frac{4}{3}, 4)$
$S' = (0, -\frac{2}{3})$
$T' = (-\frac{10}{3}, 2)$

B
$R' = (3, -9)$
$S' = (0, \frac{3}{2})$
$T' = (\frac{15}{2}, -\frac{9}{2})$

C
$R' = (-5, 15)$
$S' = (0, -\frac{5}{2})$
$T' = (-\frac{25}{2}, \frac{15}{2})$

D
$R' = (-3, 9)$
$S' = (0, -\frac{3}{2})$
$T' = (-\frac{15}{2}, \frac{9}{2})$

01 / CHOOSE & CHECK

Q10

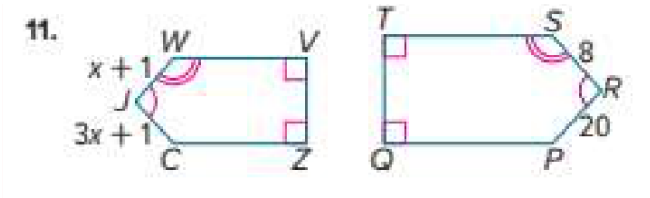
Q10 Similar polygons
المضلعات المتشابهة

Remember Corresponding angles of similar polygons are congruent. Corresponding side lengths have one common ratio. Match vertices in the stated order.

10.1 MCQ

Each pair of polygons is similar. Find x.

Reveal · p. 128 · 11 · MCQ adaptation

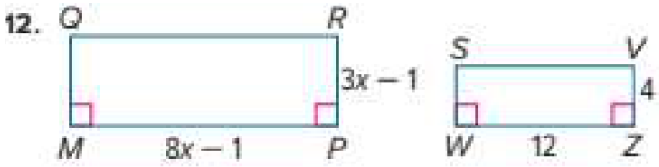


A 6
B 3
C 4
D 2

10.2 MCQ

Each pair of polygons is similar. Find x.

Reveal · p. 128 · 12 · MCQ adaptation

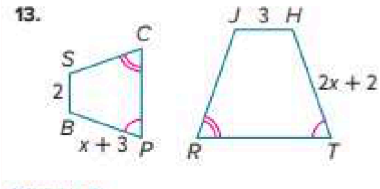


A 2
B 3
C 1
D 5

10.3 MCQ

Each pair of polygons is similar. Find x.

Reveal · p. 128 · 13 · MCQ adaptation

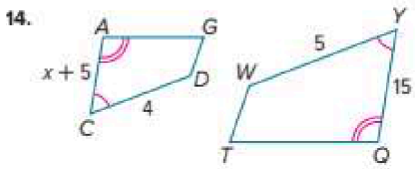


A 6
B 4
C 8
D 5

10.4 MCQ

Each pair of polygons is similar. Find x.

Reveal · p. 128 · 14 · MCQ adaptation



A 6
B 10
C 7
D 8

01 / CHOOSE & CHECK

Q11

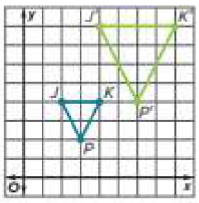
Q11 Dilation scale factors
معاملات التمدد

Remember Scale factor = image length/original length. For positive k, 0 < k < 1 gives a reduction and k > 1 gives an enlargement. A dilation preserves angle measures.

11.1 MCQ

Find the scale factor from the original polygon to its primed image.

Reveal · p. 120 · 10 · MCQ adaptation



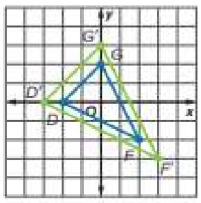
A 2 B $\frac{1}{2}$

C -2 D 3

11.2 MCQ

Find the scale factor from the original polygon to its primed image.

Reveal · p. 120 · 11 · MCQ adaptation



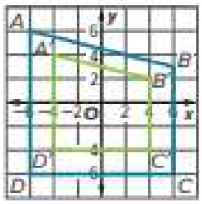
A $\frac{2}{3}$ B $-\frac{3}{2}$

C $\frac{5}{2}$ D $\frac{3}{2}$

11.3 MCQ

Find the scale factor from the original polygon to its primed image.

Reveal · p. 120 · 17 · MCQ adaptation



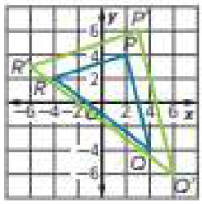
A $-\frac{2}{3}$ B $\frac{5}{3}$

C $\frac{2}{3}$ D $\frac{3}{2}$

11.4 MCQ

Find the scale factor from the original polygon to its primed image.

Reveal · p. 120 · 18 · MCQ adaptation



A $\frac{5}{2}$ B $\frac{3}{2}$

C $\frac{2}{3}$ D $-\frac{3}{2}$

01 / CHOOSE & CHECK

Q12

Q12 AA triangle similarity

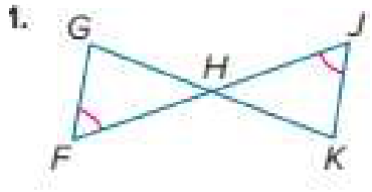
تشابه المثلثات بزاويتين

Remember Two pairs of congruent corresponding angles establish triangle similarity. Parallel lines and vertical angles can supply those angle pairs.

12.1 MCQ

Determine whether the triangles are similar. Choose a valid explanation.

Reveal · p. 133 · 1 · MCQ adaptation



A Not similar: equal vertical angles prevent similarity.

B Not enough information: no angle pair is known.

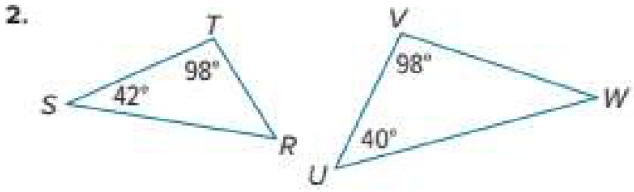
C Similar by SSS: three side lengths are given.

D Similar by AA: the marked angles and vertical angles are equal.

12.2 MCQ

Determine whether the triangles are similar. Choose a valid explanation.

Reveal · p. 133 · 2 · MCQ adaptation



A Not enough information: the third angles cannot be found.

B Similar by SSS only.

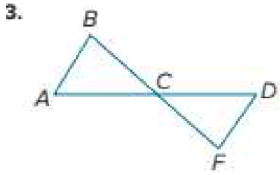
C Similar by AA: their angle sets are 42°, 98°, 40°.

D Not similar: one triangle has 42° and the other has 40°.

12.3 MCQ

Determine whether the triangles are similar. Choose a valid explanation.

Reveal · p. 133 · 3 · MCQ adaptation



A Definitely not similar: vertical angles have different measures.

B Not enough information: only the vertical angle pair is known.

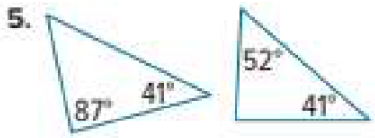
C Similar by AA: a second pair is marked congruent.

D Similar by SSS: all side ratios are given.

12.4 MCQ

Determine whether the triangles are similar. Choose a valid explanation.

Reveal · p. 133 · 5 · MCQ adaptation



A Similar by AA: both triangles have angles 87°, 41°, 52°.

B Not similar: the displayed angle values differ.

C Not enough information: a third angle cannot be calculated.

D Similar by a scale factor of 2 shown on the sides.

01 / CHOOSE & CHECK

Q13

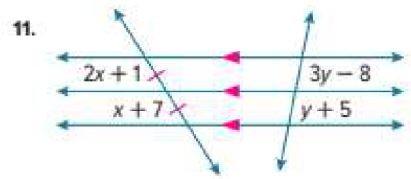
Q13 Proportional parts in triangles
الأجزاء المتناسبة في المثلثات

Remember A line parallel to one side of a triangle divides the other two sides proportionally. A triangle midsegment is parallel to the third side and half its length.

13.1 MCQ

Find the labelled variable(s), using the parallel-line and midpoint markings.

Reveal · p. 146 · 11 · MCQ adaptation

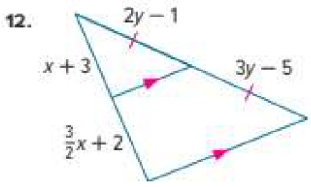


A $(x, y) = (7, 6)$
B $(x, y) = (13, 6)$
C $(x, y) = (6, \frac{13}{2})$
D $(x, y) = (6, 13)$

13.2 MCQ

Find the labelled variable(s), using the parallel-line and midpoint markings.

Reveal · p. 146 · 12 · MCQ adaptation

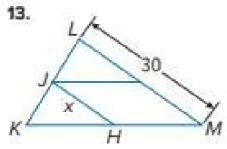


A $(x, y) = (3, 4)$
B $(x, y) = (2, 4)$
C $(x, y) = (4, 2)$
D $(x, y) = (2, 3)$

13.3 MCQ

Find the labelled variable(s), using the parallel-line and midpoint markings.

Reveal · p. 146 · 13 · MCQ adaptation

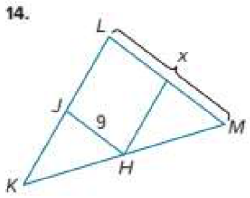


A $x = 15$
B $x = 30$
C $x = 60$
D $x = 20$

13.4 MCQ

Find the labelled variable(s), using the parallel-line and midpoint markings.

Reveal · p. 146 · 14 · MCQ adaptation



A $x = 9$
B $x = 4.5$
C $x = 27$
D $x = 18$

01 / CHOOSE & CHECK

Q14

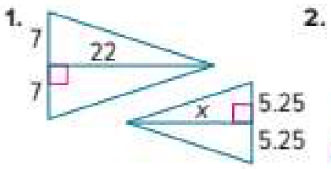
Q14 Parts of similar triangles
أجزاء المثلثات المتشابهة

Remember Corresponding altitudes, medians, and angle-bisector lengths of similar triangles have the same ratio as corresponding sides. Within one triangle, an angle bisector divides the opposite side in the ratio of the adjacent sides.

14.1 MCQ

The triangles are similar. Find x.

Reveal · p. 153 · 1 · MCQ adaptation

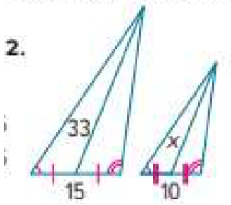


A $\frac{35}{2}$
B $\frac{33}{2}$
C 33
D $\frac{33}{4}$

14.2 MCQ

The triangles are similar. Find x.

Reveal · p. 153 · 2 · MCQ adaptation

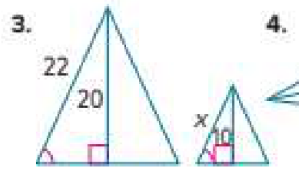


A 22
B 44
C 11
D 23

14.3 MCQ

The triangles are similar. Find x.

Reveal · p. 153 · 3 · MCQ adaptation

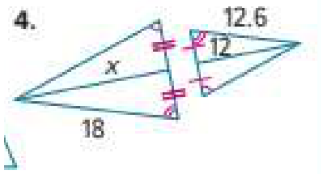


A 22
B $\frac{11}{2}$
C 12
D 11

14.4 MCQ

The triangles are similar. Find x.

Reveal · p. 153 · 4 · MCQ adaptation



A $\frac{60}{7}$
B $\frac{127}{7}$
C $\frac{120}{7}$
D $\frac{240}{7}$

01 / CHOOSE & CHECK

Q15

Q15 Geometric mean
الوسط الهندسي

Remember The geometric mean of positive a and b is √(ab). For a right triangle with altitude h to a hypotenuse split into p and q, $h^2 = pq$. A leg squared equals the whole hypotenuse times its adjacent hypotenuse segment.

15.1 MCQ

Find the geometric mean between the two numbers.

$4 \quad \text{and} \quad 6$

Reveal · p. 165 · 1 · MCQ adaptation

A $2\sqrt{6}$

B $-2\sqrt{6}$

C $4\sqrt{6}$

D $3 + 2\sqrt{6}$

15.2 MCQ

Find the geometric mean between the two numbers.

$\frac{1}{2} \quad \text{and} \quad 2$

Reveal · p. 165 · 2 · MCQ adaptation

A -1

B 2

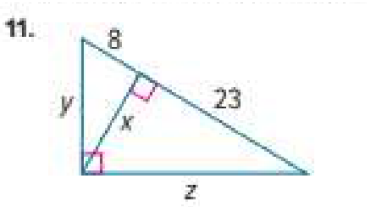
C 4

D 1

15.3 MCQ

Find all of x, y and z in simplest exact form.

Reveal · p. 165 · 11 · MCQ adaptation



A $(x, y, z) = (\sqrt{184}, \sqrt{713}, \sqrt{248})$

B $(x, y, z) = (\sqrt{248}, \sqrt{184}, \sqrt{713})$

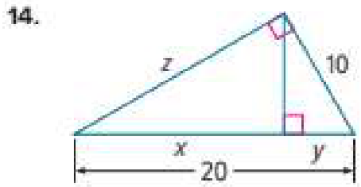
C $(x, y, z) = (2\sqrt{46}, 2\sqrt{62}, \sqrt{713})$

D $(x, y, z) = (\sqrt{31}, \sqrt{184}, \sqrt{248})$

15.4 MCQ

Find all of x, y and z in simplest exact form.

Reveal · p. 165 · 14 · MCQ adaptation



A $(x, y, z) = (10, 10, 10\sqrt{2})$

B $(x, y, z) = (15, 5, 10\sqrt{3})$

C $(x, y, z) = (5, 15, 10\sqrt{3})$

D $(x, y, z) = (15, 5, 5\sqrt{3})$

01 / CHOOSE & CHECK

Q16

Q16 Pythagorean theorem and converse

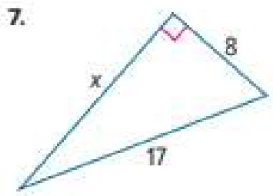
نظرية فيثاغورس وعكسها

Remember For a right triangle, $a^2 + b^2 = c^2$ with c the hypotenuse. For valid side lengths and longest side c, compare c^2 with $a^2 + b^2$ to classify the triangle.

16.1 MCQ

Use a Pythagorean triple to find x.

Reveal · p. 171 · 7 · MCQ adaptation



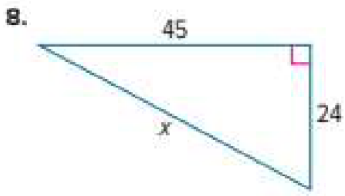
A 17 B 13

C 20 D 15

16.2 MCQ

Use a Pythagorean triple to find x.

Reveal · p. 171 · 8 · MCQ adaptation



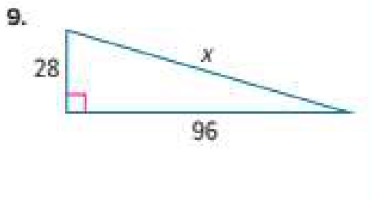
A 49 B 56

C 51 D 53

16.3 MCQ

Use a Pythagorean triple to find x.

Reveal · p. 171 · 9 · MCQ adaptation



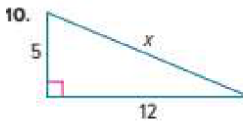
A 105 B 100

C 102 D 98

16.4 MCQ

Use a Pythagorean triple to find x.

Reveal · p. 171 · 10 · MCQ adaptation



A 13 B 15

C 11 D 18

01 / CHOOSE & CHECK

Q17

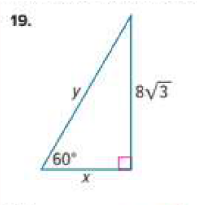
Q17 Special right triangles
المثلثات القائمة الخاصة

Remember 45°–45°–90° side ratio is 1:1:√2. In a 30°–60°–90° triangle, sides opposite 30°, 60°, and 90° have ratio 1:√3:2.

17.1 MCQ

Find x and y in simplest exact form.

Reveal · p. 184 · 19 · MCQ adaptation



A $(x, y) = (8\sqrt{3}, 16)$

B $(x, y) = (8, 8\sqrt{3})$

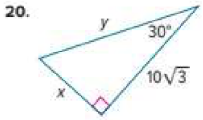
C $(x, y) = (8, 16)$

D $(x, y) = (16, 8)$

17.2 MCQ

Find x and y in simplest exact form.

Reveal · p. 184 · 20 · MCQ adaptation



A $(x, y) = (10, 10\sqrt{3})$

B $(x, y) = (10, 20)$

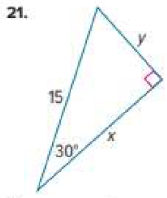
C $(x, y) = (20, 10)$

D $(x, y) = (10\sqrt{3}, 20)$

17.3 MCQ

Find x and y in simplest exact form.

Reveal · p. 184 · 21 · MCQ adaptation



A $(x, y) = (\frac{15\sqrt{3}}{2}, \frac{15}{2})$

B $(x, y) = (\frac{15}{2}, \frac{15\sqrt{3}}{2})$

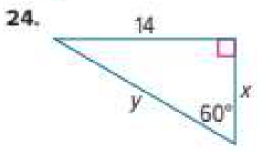
C $(x, y) = (15\sqrt{3}, 15)$

D $(x, y) = (\frac{15\sqrt{2}}{2}, \frac{15}{2})$

17.4 MCQ

Find x and y in simplest exact form.

Reveal · p. 184 · 24 · MCQ adaptation



A $(x, y) = (14, 28)$

B $(x, y) = (14\sqrt{3}, 28\sqrt{3})$

C $(x, y) = (\frac{28\sqrt{3}}{3}, \frac{14\sqrt{3}}{3})$

D $(x, y) = (\frac{14\sqrt{3}}{3}, \frac{28\sqrt{3}}{3})$

01 / CHOOSE & CHECK

Q18

Q18 Right-triangle trigonometry

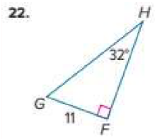
النسب المثلثية في المثلث القائم

Remember sin θ = opposite/hypotenuse, cos θ = adjacent/hypotenuse, tan θ = opposite/adjacent. Use inverse ratios for angles. Set a scientific calculator to degrees.

18.1 MCQ

Solve the right triangle. Round sides to the nearest tenth and acute angles to the nearest degree.

Reveal · p. 192 · 22 · MCQ adaptation



$\angle G = 58^\circ$

A $FH \approx 6.9$
$GH \approx 13.0$
$\angle G = 58^\circ$

B $FH \approx 17.6$
$GH \approx 20.8$
$\angle G = 58^\circ$

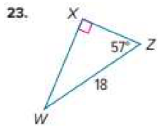
C $FH \approx 20.8$
$GH \approx 17.6$
$\angle G = 32^\circ$

D $FH \approx 17.6$
$GH \approx 20.8$

18.2 MCQ

Solve the right triangle. Round sides to the nearest tenth and acute angles to the nearest degree.

Reveal · p. 192 · 23 · MCQ adaptation



$\angle W = 33^\circ$

A $WX \approx 15.1$
$XZ \approx 9.8$
$\angle W = 33^\circ$

B $WX \approx 9.8$
$XZ \approx 15.1$
$\angle W = 57^\circ$

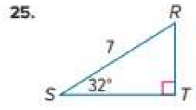
C $WX \approx 15.1$
$XZ \approx 9.8$
$\angle W = 33^\circ$

D $WX \approx 21.5$
$XZ \approx 33.1$

18.3 MCQ

Solve the right triangle. Round sides to the nearest tenth and acute angles to the nearest degree.

Reveal · p. 192 · 25 · MCQ adaptation



$\angle R = 58^\circ$

A $ST \approx 3.7$
$RT \approx 5.9$
$\angle R = 32^\circ$

B $ST \approx 5.9$
$RT \approx 3.7$
$\angle R = 58^\circ$

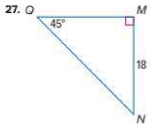
C $ST \approx 8.3$
$RT \approx 13.2$
$\angle R = 58^\circ$

D $ST \approx 5.9$
$RT \approx 3.7$

18.4 MCQ

Solve the right triangle. Round sides to the nearest tenth and acute angles to the nearest degree.

Reveal · p. 192 · 27 · MCQ adaptation



$\angle N = 45^\circ$

A $QM \approx 25.5$
$QN = 18$
$\angle N = 90^\circ$

B $QM = 18$
$QN \approx 25.5$
$\angle N = 45^\circ$

C $QM = 18$
$QN \approx 25.5$
$\angle N = 45^\circ$

D $QM = 18$
$QN = 36$

01 / CHOOSE & CHECK

Q19

Q19 Angles of elevation and depression

زوايا الارتفاع والانخفاض

Remember Angles of elevation and depression are measured from horizontal lines. Use vertical height differences, including eye height when given.

19.1 MCQ

A lighthouse light is 30 m above sea level. Sailors see it at a 25° angle of elevation. How far is the ship from shore, to the nearest metre?

Reveal · p. 199 · 1 · MCQ adaptation

A 64
B 65
C 63
D 69

19.2 MCQ

A water tower is 110 ft from a student and is 32.5 ft high. The student's eye level is 6 ft. Find the angle of elevation to the nearest tenth of a degree.

Reveal · p. 199 · 2 · MCQ adaptation

A $\frac{29}{2}$
B $\frac{25}{2}$
C $\frac{37}{2}$
D $\frac{27}{2}$

19.3 MCQ

A ladder reaches a 30-ft roof and makes a 55° angle with the ground. How far is its foot from the wall, to the nearest foot?

Reveal · p. 199 · 3 · MCQ adaptation

A 20
B 26
C 21
D 22

19.4 MCQ

Nabuko is 60 m from a cliff with bridges 100 m high. Her line of sight is 5 m above the ground. Find the elevation angle to the nearest tenth of a degree.

Reveal · p. 199 · 4 · MCQ adaptation

A $\frac{627}{10}$
B $\frac{577}{10}$
C $\frac{587}{10}$
D $\frac{567}{10}$

01 / CHOOSE & CHECK

Q20

Q20 Area with two sides and an angle

المساحة بضلعين والزاوية المحصورة

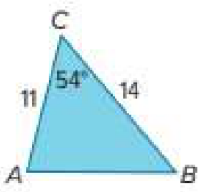
Remember Area = ½ab sin C, where C is the included angle between sides a and b. If two other angles are given, find the included angle using the 180° sum.

20.1 MCQ

Use trigonometry to find the triangle's area to the nearest tenth.

Reveal · p. 200 · 13 · MCQ adaptation

13.



A $\frac{633}{10}$

B $\frac{643}{10}$

C $\frac{623}{5}$

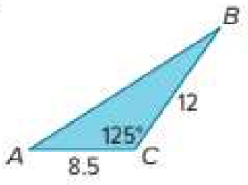
D $\frac{623}{10}$

20.2 MCQ

Use trigonometry to find the triangle's area to the nearest tenth.

Reveal · p. 200 · 14 · MCQ adaptation

14.



A $\frac{219}{5}$

B $\frac{418}{5}$

C $\frac{209}{5}$

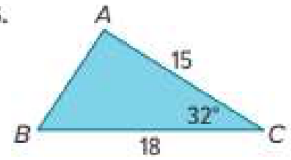
D $\frac{214}{5}$

20.3 MCQ

Use trigonometry to find the triangle's area to the nearest tenth.

Reveal · p. 200 · 15 · MCQ adaptation

15.



A 143

B $\frac{143}{2}$

C $\frac{145}{2}$

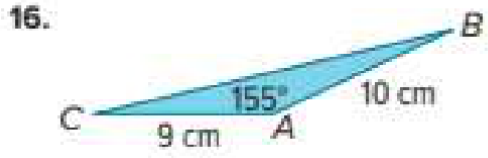
D $\frac{147}{2}$

20.4 MCQ

Use trigonometry to find the triangle's area to the nearest tenth.

Reveal · p. 200 · 16 · MCQ adaptation

16.



A 19

B 20

C 21

D 38

02 / WRITE THE STEPS

Q21 · Written practice 1

Q21

Special-product factorization

التحليل باستعمال المتطابقات الخاصة

Remember $A^2 - B^2 = (A - B)(A + B)$. $A^2 \pm 2AB + B^2 = (A \pm B)^2$. Extract the GCF before applying a special-product pattern.

21.1 FRQ · show your working

Factor completely using integers.

$$-72p^2 + 8x^2$$

Reveal · p. 680 · 35

(a) Show your factorisation and check by expansion.

21.2 FRQ · show your working

Factor completely using integers.

$$16b^2 - 100$$

Reveal · p. 680 · 38

(a) Show your factorisation and check by expansion.

21.3 FRQ · show your working

Determine whether this is a perfect-square trinomial; if so, factor it.

$$9z^2 - 6z + 1$$

Reveal · p. 681 · 51

(a) Show your factorisation and check by expansion.

21.4 FRQ · show your working

Determine whether this is a perfect-square trinomial; if so, factor it.

$$36x^2 - 60x + 25$$

Reveal · p. 681 · 52

(a) Show your factorisation and check by expansion.

02 / WRITE THE STEPS

Q22 · Written practice 1

Q22

Interior and exterior angle reasoning

الاستدلال بالزوايا الداخلية والخارجية

Remember For a convex polygon, diagonals from one vertex form n − 2 triangles. Interior sum = (n − 2)180°. One exterior angle at each vertex, taken consistently around the polygon, sums to 360°.

22.1 FRQ · show your working

The given angle is an interior angle of a regular polygon. Find the number of sides.

$144°$

Reveal · p. 64 · 9

(a) Show an equation and solve for n.

22.2 FRQ · show your working

The given angle is an interior angle of a regular polygon. Find the number of sides.

$160°$

Reveal · p. 64 · 11

(a) Show an equation and solve for n.

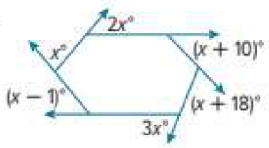
22.3 FRQ · show your working

Find x in the diagram.

Reveal · p. 64 · 15

(a) Show how you use the exterior-angle sum.

15.



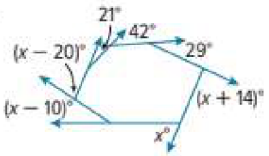
22.4 FRQ · show your working

Find x in the diagram.

Reveal · p. 64 · 18

(a) Show how you use the exterior-angle sum.

18.



02 / WRITE THE STEPS

Q23 · Written practice 1

Q23 Solving with similar triangles

حل المسائل باستعمال المثلثات المتشابهة

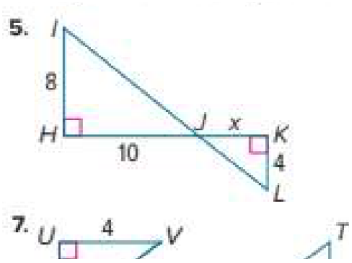
Remember Establish similarity by AA, SAS, or SSS. Write corresponding vertices in order, set up a proportion, and solve. Check that all lengths use the same units.

23.1 FRQ · show your working

Identify the similar triangles. Then find x.

Reveal · p. 139 · 5

(a) State corresponding vertices and show a proportion.

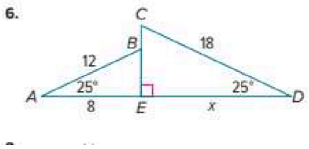


23.2 FRQ · show your working

Identify the similar triangles. Then find x.

Reveal · p. 139 · 6

(a) State corresponding vertices and show a proportion.



02 / WRITE THE STEPS

Q23 · Written practice 2

Q23 Solving with similar triangles

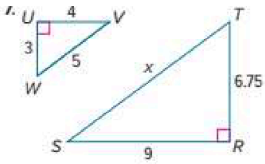
حل المسائل باستعمال المثلثات المتشابهة

Remember Establish similarity by AA, SAS, or SSS. Write corresponding vertices in order, set up a proportion, and solve. Check that all lengths use the same units.

23.3 FRQ · show your working

Identify the similar triangles. Then find x.

Reveal · p. 139 · 7

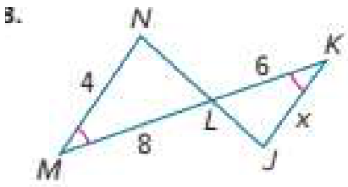


(a) State corresponding vertices and show a proportion.

23.4 FRQ · show your working

Identify the similar triangles. Then find x.

Reveal · p. 139 · 8



(a) State corresponding vertices and show a proportion.

02 / WRITE THE STEPS

Q24 · Written practice 1

Q24

Distance and midpoint in space

المسافة ومنتصف القطعة في الفراغ

Remember Distance = $\sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2]}$. Midpoint = $((x_1 + x_2)/2, (y_1 + y_2)/2, (z_1 + z_2)/2)$.

24.1 FRQ · show your working

Find the midpoint of the segment joining the two points.

$$K(-2, -4, -4), \quad L(4, 2, 0)$$

Reveal · p. 177 · 13

(a) Determine the coordinates of midpoint M.

24.2 FRQ · show your working

Find the midpoint of the segment joining the two points.

$$R(3, 3, 4), \quad V(5, 4, 13)$$

Reveal · p. 177 · 15

(a) Determine the coordinates of midpoint M.

02 / WRITE THE STEPS

Q24 · Written practice 2

Q24

Distance and midpoint in space

المسافة ومنتصف القطعة في الفراغ

Remember Distance = $\sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2]}$. Midpoint = $((x_1 + x_2)/2, (y_1 + y_2)/2, (z_1 + z_2)/2)$.

24.3 FRQ · show your working

Find the distance and midpoint of the segment.

$P(-5, -2, -1), \quad Q(-1, 0, 3)$

Reveal · p. 178 · 19

(a) Determine the distance.

(b) Determine the coordinates of midpoint M.

24.4 FRQ · show your working

Find the distance and midpoint of the segment.

$B(\sqrt{3}, 2, 2\sqrt{2}), \quad C(-2\sqrt{3}, 4, 4\sqrt{2})$

Reveal · p. 178 · 23

(a) Determine the distance.

(b) Determine the coordinates of midpoint M.

02 / WRITE THE STEPS

Q25 · Written practice 1

Q25

Law of Sines and the ambiguous case

قانون الجيوب والحالة المبهمة

Remember Use a/sin A = b/sin B. For SSA, sin B = b sin A/a. Test both $B_1 = \sin^{-1}(\text{value})$ and $B_2 = 180° - B_1$, accepting only cases with A + B < 180°. Opposite sides and angles share letters.

25.1 FRQ · show your working

Determine whether the triangle has no solution, one solution or two solutions. Then solve it. Round sides to the nearest tenth and angles to the nearest degree.

$$A = 50°, \quad a = 34, \quad b = 40$$

Reveal · p. 208 · 15

(a) Determine the number of solutions and justify it.

(b) Give every possible triangle, or explain why none exists.

25.2 FRQ · show your working

Determine whether the triangle has no solution, one solution or two solutions. Then solve it. Round sides to the nearest tenth and angles to the nearest degree.

$$A = 24°, \quad a = 3, \quad b = 8$$

Reveal · p. 208 · 16

(a) Determine the number of solutions and justify it.

(b) Give every possible triangle, or explain why none exists.

02 / WRITE THE STEPS

Q25 · Written practice 2

Q25

Law of Sines and the ambiguous case

قانون الجيوب والحالة المبهمة

Remember Use a/sin A = b/sin B. For SSA, sin B = b sin A/a. Test both $B_1 = \sin^{-1}(\text{value})$ and $B_2 = 180° - B_1$, accepting only cases with A + B < 180°. Opposite sides and angles share letters.

25.3 FRQ · show your working

Determine whether the triangle has no solution, one solution or two solutions. Then solve it. Round sides to the nearest tenth and angles to the nearest degree.

$$A = 125°, \quad a = 22, \quad b = 15$$

Reveal · p. 208 · 17

(a) Determine the number of solutions and justify it.

(b) Give every possible triangle, or explain why none exists.

25.4 FRQ · show your working

Determine whether the triangle has no solution, one solution or two solutions. Then solve it. Round sides to the nearest tenth and angles to the nearest degree.

$$A = 30°, \quad a = 2, \quad b = 4$$

Reveal · p. 208 · 19

(a) Determine the number of solutions and justify it.

(b) Give every possible triangle, or explain why none exists.

03 / CHECK & LEARN

MCQ answer key

Check after solving. Match the printed question number to its correct option. Full explanations are available in the online LO bank.

Q1 · Polynomial operations
1.1: C 1.2: B 1.3: A 1.4: D

Q2 · FOIL and binomial products
2.1: B 2.2: A 2.3: D 2.4: C

Q3 · Factoring by GCF and grouping
3.1: A 3.2: D 3.3: C 3.4: B

Q4 · Factoring monic trinomials
4.1: D 4.2: C 4.3: B 4.4: A

Q5 · Interior angles of polygons
5.1: C 5.2: B 5.3: A 5.4: D

Q6 · Properties of parallelograms
6.1: B 6.2: A 6.3: D 6.4: C

Q7 · Tests for parallelograms
7.1: A 7.2: D 7.3: C 7.4: B

Q8 · Kites
8.1: D 8.2: C 8.3: B 8.4: A

Q9 · Dilation coordinates
9.1: C 9.2: B 9.3: A 9.4: D

Q10 · Similar polygons
10.1: B 10.2: A 10.3: D 10.4: C

Q11 · Dilation scale factors
11.1: A 11.2: D 11.3: C 11.4: B

Q12 · AA triangle similarity
12.1: D 12.2: C 12.3: B 12.4: A

Q13 · Proportional parts in triangles
13.1: C 13.2: B 13.3: A 13.4: D

Q14 · Parts of similar triangles
14.1: B 14.2: A 14.3: D 14.4: C

Q15 · Geometric mean
15.1: A 15.2: D 15.3: C 15.4: B

Q16 · Pythagorean theorem and converse
16.1: D 16.2: C 16.3: B 16.4: A

Q17 · Special right triangles
17.1: C 17.2: B 17.3: A 17.4: D

Q18 · Right-triangle trigonometry
18.1: B 18.2: A 18.3: D 18.4: C

Q19 · Angles of elevation and depression
19.1: A 19.2: D 19.3: C 19.4: B

Q20 · Area with two sides and an angle
20.1: D 20.2: C 20.3: B 20.4: A

03 / CHECK & LEARN

Worked solutions 1

21.1 p. 680 · 35

(a) Extract any greatest common factor, then use the difference-of-squares or perfect-square identity. Correct method (5), complete factors and check (5).

$$8\left(-3p+x\right)\left(3p+x\right)$$

21.2 p. 680 · 38

(a) Extract any greatest common factor, then use the difference-of-squares or perfect-square identity. Correct method (5), complete factors and check (5).

$$4\left(2b-5\right)\left(2b+5\right)$$

21.3 p. 681 · 51

(a) Yes; Extract any greatest common factor, then use the difference-of-squares or perfect-square identity. Correct method (5), complete factors and check (5).

$$\left(3z-1\right)^{2}$$

21.4 p. 681 · 52

(a) Yes; Extract any greatest common factor, then use the difference-of-squares or perfect-square identity. Correct method (5), complete factors and check (5).

$$\left(6x-5\right)^{2}$$

22.1 p. 64 · 9

(a) Exterior angle=180−144=36°. Therefore n=360/36=10. Correct exterior angle (3), equation (3), number of sides (4).

$$n=10$$

22.2 p. 64 · 11

(a) Exterior angle=180−160=20°. Therefore n=360/20=18. Correct exterior angle (3), equation (3), number of sides (4).

$$n=18$$

22.3 p. 64 · 15

(a) The exterior angles sum to 360° (3). Combine all labelled exterior angles: 9x+27=360 (4), so x=37 (3).

$$x=37$$

22.4 p. 64 · 18

(a) The exterior angles sum to 360° (3). Combine all labelled exterior angles: 4x+76=360 (4), so x=71 (3).

$$x=71$$

23.1 p. 139 · 5

(a) The vertical angles at J are equal and H,K are right angles. 8/4=10/x gives x=5. Correct ordered correspondence (4), proportion (3), x (3).

$$\triangle IHJ\sim\triangle LKJ,\quad x=5$$

23.2 p. 139 · 6

(a) Both have a right angle at E and a marked 25° angle. 12/18=8/x gives x=12. Correct ordered correspondence (4), proportion (3), x (3).

$$\triangle ABE\sim\triangle DCE,\quad x=12$$

03 / CHECK & LEARN

Worked solutions 2

23.3 p. 139 · 7

(a) The ratios 9/4=6.75/3=2.25 agree. The hypotenuse x=5·2.25=11.25. Correct ordered correspondence (4), proportion (3), x (3).

$$\triangle UVW \sim \triangle RTS, \quad x = \frac{45}{4}$$

23.4 p. 139 · 8

(a) Use the marked angles at M,K and vertical angles at L. 4/x=8/6 gives x=3. Correct ordered correspondence (4), proportion (3), x (3).

$$\triangle MNL \sim \triangle KJL, \quad x = 3$$

24.1 p. 177 · 13

(a) Average the corresponding x, y and z coordinates separately. Correct component calculations and complete midpoint.

$$M = (1, -1, -2)$$

24.2 p. 177 · 15

(a) Average the corresponding x, y and z coordinates separately. Correct component calculations and complete midpoint.

$$M = (4, \tfrac{7}{2}, \tfrac{17}{2})$$

24.3 p. 178 · 19

(a) Use $d=\sqrt{[(x_2-x_1)^2+(y_2-y_1)^2+(z_2-z_1)^2]}$; substitution (3) and simplified exact distance (2).

$$d = 6$$

(b) Average the corresponding x, y and z coordinates separately. Correct component calculations and complete midpoint.

$$M = (-3, -1, 1)$$

24.4 p. 178 · 23

(a) Use $d=\sqrt{[(x_2-x_1)^2+(y_2-y_1)^2+(z_2-z_1)^2]}$; substitution (3) and simplified exact distance (2).

$$d = \sqrt{39}$$

(b) Average the corresponding x, y and z coordinates separately. Correct component calculations and complete midpoint.

$$M = (-\tfrac{\sqrt{3}}{2}, 3, 3\sqrt{2})$$

25.1 p. 208 · 15

(a) sin B=b sin A/a=0.90122876. Two solutions; test both arcsine candidates against A+B<180°. Correct ratio (2), count and admissibility (2).

(b) B≈64°, C≈66°, c≈40.4; B≈116°, C≈14°, c≈11.0 Calculate with full-precision angles before rounding. Correct possible triangles and remaining side(s) (6).

25.2 p. 208 · 16

(a) sin B=b sin A/a=1.08463105. No solution; test both arcsine candidates against A+B<180°. Correct ratio (2), count and admissibility (2).

(b) sin B>1, so there is no real angle B and no triangle. Calculate with full-precision angles before rounding. Correct possible triangles and remaining side(s) (6).

25.3 p. 208 · 17

(a) sin B=b sin A/a=0.55851276. One solution; test both arcsine candidates against A+B<180°. Correct ratio (2), count and admissibility (2).

(b) B≈34°, C≈21°, c≈9.6 Calculate with full-precision angles before rounding. Correct possible triangles and remaining side(s) (6).

25.4 p. 208 · 19

(a) sin B=b sin A/a=1.00000000. Two solutions; test both arcsine candidates against A+B<180°. Correct ratio (2), count and admissibility (2).

(b) B≈90°, C≈60°, c≈3.5; B≈90°, C≈60°, c≈3.5 Calculate with full-precision angles before rounding. Correct possible triangles and remaining side(s) (6).